Huanxuan Liao

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Education

Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing 2023 – Now

PhD in Computer Science. In NLPR, Supervisor: Shizhu He and Kang Liu

North China Electric Power University (NCEPU), Beijing 2019 – 2023

Bachelor of Engineering in Intelligence Science and Technology (Innovation Class). Ranking 1/13

Research Interests

- **Efficiency Optimization for Long Sequence & Multimodal**:
  - Survey: Awesome-LLM-Long-Context-Modeling (GitHub Stars > 2.2K)
  - KV Cache Compression: SparK (AAAI 2026)
  - Hybrid Context Compression: HyCo2 (arXiv 2025)
  - Dynamic RAG Efficiency: DyPRAG (arXiv 2025)
  - Episodic Memory Internalization: SKIntern (COLING 2025)
  - Multimodal Efficiency: ResAdapt (arXiv 2026)
- **Reasoning Capability & Policy Optimization**:
  - Instruction Learning: TAGI (NeurIPS 2024)
  - Instruction-Parameter Shuttle: SHIP (ACL 2026 Main)
  - Parametric Knowledge Activation: AAG (COLING 2025)
  - Neural-Symbolic Collaboration: NesyCD (AAAI 2025)
  - Continual Knowledge Transfer: SpaRTA (ACL 2026 Main), PaRSP (ACL 2026 Main)
  - Policy Optimization: BuPO (arXiv 2025)
- **Agentic System & Tool Learning**:
  - Tool Learning: CITI (AAAI 2025)
  - Tool Benchmark: ETAPP (ACL 2025 Main)
  - Tool-Use RL: KATE (ACL 2026 Findings), Tool-RL-Box (FAGEN@ICML 2026)
  - Agent Memory: Distill Yourself

Selected Publications [full list]

(* stands for equal contribution; Listed in reverse chronological order.)

- **Huanxuan Liao**, Zhongtao Jiang, Yupu Hao, Yuqiao Tan, Shizhu He, Ben Wang, Jun Zhao, Kun Xu, Kang Liu. ResAdapt: Adaptive Resolution for Efficient Multimodal Reasoning. (arXiv 2026)

- **Huanxuan Liao**, Shizhu He, Yupu Hao, Yequan Wang, Wenhao Teng, Xiangwen Liao, Jun Zhao, Kang Liu. Spectral Disentanglement: Rank-Aware Task Adaptation for Rehearsal-free Continual Learning in LLMs. (ACL 2026 Main, CCF-A)

- **Huanxuan Liao**, Yixing Xu, Shizhu He, Guanchen Li, Xuanwu Yin, Dong Li, Emad Barsoum, Jun Zhao, Kang Liu. SparK: Query-Aware Unstructured Sparsity with Recoverable KV Cache Channel Pruning. (AAAI 2026, CCF-A)

- **Huanxuan Liao**, Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Jun Zhao. Neural-Symbolic Collaborative Distillation: Advancing Small Language Models for Complex Reasoning Tasks. (AAAI 2025, CCF-A)

- **Huanxuan Liao**, Shizhu He, Yupu Hao, Xiang Li, Yuanzhe Zhang, Jun Zhao, Kang Liu. SKIntern: Internalizing Symbolic Knowledge for Distilling Better CoT Capabilities into Small Language Models. (COLING 2025, CCF-B)

- **Huanxuan Liao**, Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Shengping Liu, Jun Zhao. Awakening Augmented Generation: Learning to Awaken Internal Knowledge of Large Language Models for Question Answering. (COLING 2025, CCF-B)

- **Huanxuan Liao**, Shizhu He, Yao Xu, Yuanzhe Zhang, Yanchao Hao, Shengping Liu, Kang Liu, Jun Zhao. From Instance Training to Instruction Learning: Task Adapters Generation from Instructions. (NeurIPS 2024, CCF-A)

- **Huanxuan Liao**, Shizhu He, Yupu Hao, Jun Zhao, Kang Liu. Beyond Hard and Soft: Hybrid Context Compression for Balancing Local and Global Information Retention. (arXiv 2025)

- Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, **Huanxuan Liao**, et al. A Comprehensive Survey on Long Context Language Modeling. (arXiv 2025, **Core Contributor**)

Internship/Project Experience

ByteDance Beijing — Jun. 2025 – Mar 2026

Intern Research Intern (Efficient Video Understanding)

- **Input-Side Resolution Adaptation**: Proposed **ResAdapt**, a framework treating visual perception as a learnable policy optimized via **Cost-Aware Policy Optimization (CAPO)**. It reduces visual tokens by **4-9**× while achieving **SOTA accuracy** on VideoMME (**66.2%**) and enabling **16**× longer context for long-form video understanding.
- **Visual Token Compression Evaluation**: Systematically reproduced and evaluated various multimodal visual token compression algorithms. Comprehensive comparisons between custom designs and SOTA methods revealed the surprising efficacy of strong random baselines and highlighted the under-utilization of temporal video information in existing models, providing empirical guidance for future architecture design.
- **Efficient Architecture Exploration**: Investigated and trained computationally efficient LLaVA variants, incorporating techniques such as skip-layer connections and dynamic boundary compression (inspired by Concept Models) to significantly expand the Pareto frontier between multimodal reasoning efficiency and performance.

Ant Group Beijing — Feb. 2025 – May 2025

Intern Research Intern (Long Context)

- **Hybrid Context Compression**: Designed a hybrid compression architecture combining dynamic routing soft tokens (via Qformer) for global semantic retention and classifier-based pruning for local detail preservation.
- **Recoverable KV Pruning**: Developed a dynamic unstructured channel pruning mechanism for KV cache with runtime channel restoration, effectively reducing memory footprint without compromising generation quality.
- **Long Context Survey**: Authored a comprehensive survey on long context modeling, systematizing recent advances based on my widely-starred GitHub repository.

Awards and Scholarships

Scholarships & Awards

China National Scholarship *(top scholarship in China; 0.2% domestically)*, Ministry of Education	Dec. 2025
Climbing Second Prize Scholarship	Dec. 2024
Beijing Outstanding Graduate Awards, Beijing Ministry of Education	Jun. 2023
China National Scholarship *(top scholarship in China; 0.2% domestically)*, Ministry of Education	Dec. 2022
China National Scholarship *(top scholarship in China; 0.2% domestically)*, Ministry of Education	Dec. 2021
SiFang Society Scholarship, NCEPU	Dec. 2020

Competition

National Third Prize, Information Security Competition	Aug. 2022
National Excellence, College Student Innovation and Entrepreneurship Project	Dec. 2021
Beijing Third Prize, Internet +	Aug. 2021

Professional Services

Conference Reviewing: ACL ARR 2024, 2025. NeurIPS 2025. ICLR 2025. AAAI 2026.
Community: Awesome-LLM-Long-Context-Modeling GitHub Stars > 2.2K; Distill Yourself, local-first agent memory and preference distillation.